

1. FCFS Scheduling algorithm

```
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

```
[eeshayuktha@Eeshas-MacBook-Air OSLab % nano fcfs.c
[eeshayuktha@Eeshas-MacBook-Air OSLab % gcc fcfs.c -o fcfs
[eeshayuktha@Eeshas-MacBook-Air OSLab % ./fcfs
Enter number of processes: 5
Enter burst time for each process:
P1: 12
P2: 3
P3: 4
P4: 6
P5: 4

Process BT      WT      tTAT
P1      12      0       12
P2       3     12      15
P3       4     15      19
P4       6     19      25
P5       4     25      29
eeshayuktha@Eeshas-MacBook-Air OSLab %
```

2. SJF_NONpreemptive

```
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

```
leeshayuktha@Eeshas-MacBook-Air OSLab % gcc nonpsjf.c -o ns
leeshayuktha@Eeshas-MacBook-Air OSLab % ./ns
```

Enter number of processes: 5

Enter burst time:

P1: 1

P2: 2

P3: 5

P4: 16

P5: 6

Process	BT	WT	tTAT
P1	1	0	1
P2	2	1	3
P3	5	3	8
P4	6	8	14
P5	16	14	30

```
leeshayuktha@Eeshas-MacBook-Air OSLab %
```

3. Priority

```
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    // Sort according to priority
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {

            if (priority[i] > priority[j]) {

                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
```

```

    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t\t%d\t%d\n",
            i + 1, bt[i], priority[i], wt[i], tat[i]);
    }

    return 0;
}

```

```

[eeshayuktha@Eeshas-MacBook-Air OSLab % nano priority.c
[eeshayuktha@Eeshas-MacBook-Air OSLab % gcc priority.c -o p
[eeshayuktha@Eeshas-MacBook-Air OSLab % ./p
Enter number of processes: 4
Enter burst time and priority:
P1 Burst Time: 1
P1 Priority: 3
P2 Burst Time: 5
P2 Priority: 1
P3 Burst Time: 4
P3 Priority: 2
P4 Burst Time: 6
P4 Priority: 4

Process BT      Priority      WT      TAT
P2      5        1            0        5
P3      4        2            5        9
P1      1        3            9       10
P4      6        4           10       16
eeshayuktha@Eeshas-MacBook-Air OSLab %

```

4. ROUND ROBIN

```

#include <stdio.h>

int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
}

```

```

scanf("%d", &n);

printf("Enter burst time:\n");

for (i = 0; i < n; i++) {
    printf("P%d: ", i + 1);
    scanf("%d", &bt[i]);

    rem_bt[i] = bt[i];
    wt[i] = 0;
}

printf("Enter time quantum: ");
scanf("%d", &quantum);

// Round Robin
while (1) {

    done = 1;

    for (i = 0; i < n; i++) {

        if (rem_bt[i] > 0) {

            done = 0;

            if (rem_bt[i] > quantum) {
                time = time + quantum;
                rem_bt[i] = rem_bt[i] - quantum;
            }
            else {
                time = time + rem_bt[i];

                wt[i] = time - bt[i];

                rem_bt[i] = 0;
            }
        }
    }

    if (done == 1)
        break;
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

```

```

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

return 0;
}

```

```

[eeshayuktha@Eeshas-MacBook-Air OSLab % nano roundrobin.c
[eeshayuktha@Eeshas-MacBook-Air OSLab % gcc roundrobin.c -o rr
[eeshayuktha@Eeshas-MacBook-Air OSLab % ./rr
Enter number of processes: 4
Enter burst time:
P1: 2
P2: 3
P3: 1
P4: 5
Enter time quantum: 2

Process BT      WT      TAT
P1      2       0       2
P2      3       5       8
P3      1       4       5
P4      5       6      11
eeshayuktha@Eeshas-MacBook-Air OSLab %

```